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MILITARY SPECIFICATION

WELDING MACHINES, ARC, AC/DC, GENERATOR, CONSTANT CURRENT, INTERNAL COMBUSTION ENGINE DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers AC/DC, constant current, welding machines, with 100 percent duty cycle and driven by an internal combustion engine.

1.2 Classification. The welding machines shall be of the following types and sizes. The type and size machine required shall be as specified (see 6.2.1).

Type I - Air Cooled Gasoline Engine Driven
Size 200 - 200 Amperes

Type II - Liquid Cooled Gasoline Engine Driven
Size 300 - 300 Amperes

Type III - Liquid Cooled Diesel Engine Driven
Size 200 - 200 Amperes
Size 300 - 300 Amperes

Type IV - Air Cooled Diesel Engine Driven
Size 200 - 200 Amperes
Size 300 - 300 Amperes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, TN 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3431

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARD

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/IEEE 268 - Metric Practice.

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ANSI/NEMA EW 1 - Electric Arc-Welding Apparatus.

ANSI/NEMA ICS 1 - Industrial Control and Systems.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Commercial Packaging, Standard Practice for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.2 Design. The machine shall be new and one of the manufacturer's current models capable of performing the welding processes in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. The machine shall be designed and rated to conform with the requirements of NEMA Standard EW-1, class I machines.

3.2.1. Measuring systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in ANSI/IEEE 268.

3.2.1.1 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI System of Measurements.

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3.2.2 Reclaimed materials. The machine may contain reclaimed materials provided such materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and all its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the Industry.

3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the welding machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.5 Environmental protection. The machine shall be designed and constructed that, under the operating, service, transportation and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state or local statutes in effect at point of installation.

3.2.6 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible.

3.2.7 Lifting eye. The machines shall be provided with a lifting eye for handling by a crane or hoist. The eye shall be located at the top of the machine at the normal center of balance. The lifting eye shall be capable of supporting two and one-half times the weight of the machine and shall be mounted in such a manner that base or skid mounted machines will retain alignment during lifting operations.

3.2.8 Enclosure. The machine shall be contained in a weatherproof enclosure or be inherently weatherproof. The type II, III and IV machines shall have hinged or removable panels for access and operation. Design of the

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enclosure shall be such that internal parts are protected from the weather even when the enclosure is opened in the operating position. The sheet metal of the enclosure shall be not less than 0.045 inches in thickness.

3.2.9 Stability. The machine shall be designed to operate satisfactorily when inclined at an angle of 15 degrees in any direction from the normal operating position.

3.2.10 Interchangeability. To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as peening, plugging, or filling with solders or paste shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be cleaned and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function.

3.3.5 Painting. All surfaces to be painted shall be cleaned of all foreign matter. Unless otherwise sepcified (see 6.2.1), the machine shall be painted in accordance with manufacturer's standard commercial practice which shall be not less than one coat of primer and one coat of finish color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced herein.

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3.3.7 Gears. All gears shall be constructed from a material suitable for the intended purpose. All steel gears shall be machined and heat treated by a process that will impart the hardness, surface finish and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure or premature deterioration and wear. All gears shall be manufactured to definite established standards to provide for interchangeability and parts replacement.

3.4 Components.

3.4.1 Engine and accessories.

3.4.1.1 Engine. The engine for the welding machine shall be a standard commercial internal combustion engine of current design and development. The engine shall develop horsepower to operate the welding generator, with accessories, at its maximum rated load. The engine for the type I machine shall be a 4-cycle, forced air cooled gasoline engine which will operate satisfactorily on leaded or unleaded gasoline of 83 or greater octane. The engine for the type II machine shall be a 4-cycle, liquid cooled gasoline engine which will operate satisfactorily on leaded or unleaded gasoline of 83 or greater octane. The engine for the type III and type IV machine shall be a 2 or 4-cycle diesel which will operate satisfactorily on #2 grade diesel fuel. The engine for the type III machine shall be liquid cooled and the engine for the type IV machine shall be forced air cooled. All engines shall comply with Government emission standards.

3.4.1.2 Starting systems. Unless otherwise specified, the type I machine shall be either manually started by a hand crank or automatic recoil pull cord or 12 volt electrical starting system. When specified (see 6.2.1), the type I machine shall have a 12-volt electrical starting system. The type II, III, and IV machines shall have a 12-volt electrical starting system. Electrically started machines shall have adequate cranking capacity to start the engine under all conditions specified herein. Unless otherwise specified (see 6.2.1), required batteries shall be provided with each machine, and the batteries shall be the maintenance-free type or dry charged, (Electrolyte is not required).

3.4.1.3 Engine lubrication system. Type II, III and IV engines shall have a pressure type lubrication system and all engines shall have provisions to drain the oil from the crankcase. The grade and capacity of the oil supply shall be specified on the filler cap or adjacent surface. When specified (see 6.2.1), the type II, III or IV engines shall be provided with a low oil pressure shut-down device to stop the engine in the event of inadequate oil pressure.

3.4.1.4 Choke controls. The type I and II machines shall have a choke control system to aid in starting the engine.

3.4.1.5 Battery charging generator. When a 12-volt starting system is required, the machines shall have a battery charging generator or alternator and regulator of sufficient size and capacity to keep the battery charged.

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3.4.1.6 Governor. Each machine shall be provided with a governor control system capable of maintaining the engine RPM required to obtain the performance requirements specified herein.

3.4.1.7 Starting aid. The type III and IV machines shall have a cold weather starting device of either the volatile fluid type or glow plug type system. If the fluid system is used, a warning statement as to when the starting aid fluid shall and shall not be injected shall be printed on a corrosion resistant plate and secured near the starting aid control.

3.4.1.8 Air cleaner. The engines shall have an air cleaner to filter all air for the air intake system. The air cleaner shall be of the wet or dry type, designed for easy disassembly and cleaning.

3.4.1.9 Oil filter. Type II, III and IV machines shall have a replaceable cartridge type oil filter.

3.4.1.10 Cooling system. Liquid cooled engines shall be provided with thermostat controlled liquid cooling system of adequate size and design to operate at an ambient temperature between -25°F and 110°F. When specified (see 6.2.1), liquid cooled engines shall have a high temperature shutdown device to stop the engine in the event of excessive temperature.

3.4.1.11 Automatic idling device. When specified (see 6.2.1), the machine shall have an automatic idling device to reduce speed of the engine to an idle when not welding. Upon striking an arc, the idling device shall immediately allow the engine to speed up to its previous setting. The idling device shall have the means to keep the engine at a constant speed if so desired. The idling device shall have a time delay to allow the operator, when welding, to change electrode before the engine comes to idling speed.

3.4.1.12 Muffler. The machine shall have a muffler on the engine exhaust system. If the exhaust system is the vertical or upright type, it shall have a suitable rain cap on the end of the exhaust pipe to prevent water from entering.

3.4.1.13 Controls and instruments. The engine controls and instruments shall be grouped together in a separate panel, or be integral with the welding generator control panel. The controls for the cold weather starting aid may be located on the engine. Liquid cooled engines shall have a temperature gage or warning light, pressure lubricated engines shall have an oil pressure gage or warning light and engines with an electrical charging system shall have an ammeter or warning light. One light may be used for both temperature and oil pressure, if this system is standard with the manufacturer.

3.4.2 Fuel system.

3.4.2.1 Fuel tank. When operated at maximum load and rated duty cycle, the type I machine shall have a fuel capacity to operate not less than 3

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hours, the type II machine not less than 6 hours and the type III and IV machines not less than 8 hours. The tanks for type I and type II machines shall have drain plugs. Tanks for type III and type IV machines shall have drain petcocks.

3.4.2.2 Fuel gage. When specified (see 6.2.1), type II, III, and IV machines shall have a fuel gage to register the quantity of fuel in the tank.

3.4.2.3 Fuel lines. Fuel lines shall be installed free of sharp bends and loops that might cause failure from fatigue, strain, or vapor lock. The fuel lines shall be bracketed and fastened when necessary to prevent vibration. A shut-off cock shall be installed in the outlet of the tank when the tank is located to provide gravity feed.

3.4.2.4 Fuel strainer and filter. The type I and II machines shall have fuel filters. The type III and IV machines shall have means to filter the fuel before it enters the full injection pump. The filter shall be of the water separator type with a replaceable cartridge and water drain valve or petcock.

3.5 Performance. The output rating (amperes and volts) and performance capabilities of each machine shall be in accordance with NEMA-STD-EW-1 for class I machines. Each machine shall be rated at 100 percent duty cycle. The machines shall provide a satisfactory power source for producing the following welding processes:

a. AC welding with any electrodes manufactured for use with AC current.

b. AC welding with processes using gas shielding.

c. AC welding wherein high frequency is required.

d. DC welding with both straight and reverse polarity wherein a drooping arc characteristic is required.

Welding operations (b) and (c) are not required for the 200 ampere size welding machine.

3.5.1 Open ciruit voltage. The open circuit voltage of the welding machine shall be in accordance with NEMA EW-1.

3.5.2 Operating temperature. The operating temperature of the generator and all parts subject to damage by excessive temperature shall conform to the requirements of NEMA EW-1.

3.6 Welding generator, exciter, and auxiliary power supply. The generator and exciter shall have a class B insulation system or better in accordance with NEMA EW-1. Type II, III and IV machines shall have a voltmeter to indicate the output voltage and an ammeter to indicate output current of the machine during welding operation. A combination volt-ammeter may be used. The meter shall have an accuracy within three percent of full scale reading.

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3.6.1 Welding generator. The welding generator shall conform to the standards of NEMA EW-1. The generator shall be constant current, self-regulating type, with a drooping volt-ampere characteristic. The generator shall have double-sealed anti-friction bearings, and shall be coupled to the engine with a metal coupling to maintain alignment with the engine flywheel. The output terminals shall be of the stud and hex nut type. The studs shall be not less than 1/2-inch in diameter, and shall be recessed, protected, or located to prevent accidental contact. The electrode and work terminals shall be so identified. Two controls shall be provided to control and set the desired welding current range. One control or series of taps shall control the current through the entire range of the machine. Another control shall be provided for fine stepless adjustments within the settings of the current control. For type II, III and IV machines, a positive acting switch shall be provided to change the polarity of the welding current. The switch shall have its position marked for straight and reverse polarity. A remote control device to make fine adjustments at the work shall be furnished with type II, III and IV welding machines. The control cable shall have suitable connections for using longer extensions. The minimum length of the remote control cable shall be twenty-five feet.

3.6.2 Exciter. The exciter shall be integral with the welding generator, or shall be separate and mounted on the same shaft as the main welding generator. The exciter shall have fast voltage build up, and be designed to prevent polarity reversal. The commutator, or slip ring, shall be readily accessible for servicing and maintenance.

3.6.3 Auxiliary electric power supply. The machine shall have an auxiliary power supply. A NEMA standard 115-volt duplex receptacle shall be provided for lights and power tools. The circuit shall be overload protected by a circuit breaker or fuses. When specified, 230 volt AC auxiliary power supply shall be provided in addition to the 115-volt circuit. The 230-volt circuits shall have a NEMA Standard three-wire single or duplex receptacle. The KW capacity of the 230-volt circuit shall be as specified (see 6.2.1).

3.6.4 Wiring system. The type and size of wire used shall have sufficient capacity, temperature and voltage rating for the intended use. Wiring shall be secured to prevent damage by chafing or vibration. The wire terminals shall be codedc and a wiring diagram of the generator and engine shall be provided and fastened in a convenient location to the inside surface of the machine enclosure. The diagram shall have a transparent covering or coating for protection. The wiring shall comply with NEMA Standard ICS-1.

3.7 Machine mounting. Unless otherwise specified, the welding machine shall be furnished with a standard base, arranged and drilled for bolting the machine to the floor or deck. When required, the machine shall be mounted on skids or mounted on a trailer for portability in a shop or work area as specified (see 6.2.1).

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3.7.1 Over the road trailer. When specified (see 6.2.1), the machine shall be mounted on a two-wheel or four-wheel tandem trailer with tube or tubeless tires, not less than 600 x 14 size. The tires shall be standard pneumatic commercial type having a ply rating and size recommended in the tire and rim association yearbook for the gross weight of the unit. The trailer shall be designed in such a manner that, when fully loaded, it may be towed at speeds up to 50 mph on improved roads and up to 10 mph on rough terrain. When being towed at any speed up to 50 mph, the trailer shall track true with minimum side sway or hunting and shall support the complete welding machine and accessories. Road clearance to the trailer's lowest member shall not be less than eight inches. The trailer shall have resilient mounting (springs, torsion bars, or similar devices) to minimize transmission of road shock to the welding machine. When specified (see 6.2.1), a spare tire and wheel shall be provided and mounted in an accessible area on the trailer.

3.7.1.2 Provisions for accessories. Space shall be provided on over the road trailers for firmly mounting two-gas cylinders and storage of all hoses, cables and welding accessories. Gas cylinder mounting assembly(s) shall be provided to accommodate one-gas cylinder up to 9-1/4 inch diameter and one-gas cylinder up to 12-7/8 inch diameter. The mounting assembly(s) shall be positioned so as not to interfere with the control panel, hinder daily servicing, or add to the design volumn when the gas cylinders are installed.

3.7.1.3 Drawbar. The drawbar tongue shall be approximately 48 inches long and shall have a lunette type ring fabricated of one-inch round steel. The ring shall have an inside nominal diameter of three inches.

3.7.1.4 Lights reflectors and chains. Vehicle reflectors, lights, and safety chains shall be installed on the trailer in accordance with the manufacturer's standard practice but shall comply with pertinent ICC regulations for vehicles to be towed on highways.

3.7.1.5 Parking brake. The trailer shall be equipped with a parking brake sufficient to hold the fully loaded trailer on a 15-degree incline.

3.7.1.6 Landing gear. Two wheel trailers shall be equipped with a retractable or adjustable type landing gear with full swiveling bottom wheel. A height adjustment of not less than four inches above level to four inches below level position shall be provided.

3.8 Equipment.

3.8.1 Standard equipment. The machine shall be furnished with all standard equipment normally furnished by the manufacturer.

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3.8.2 Cables. One set of welding cables shall be furnished and the cables shall be fitted with an electrode holder, a ground clamp, and two cable lugs that will fit the terminals on the machine. The length of the cables shall be as specified (see 6.2.1). The cables shall be not less than 30-gauge stranding and shall be compatible with the maximum ampere rating and duty cycle of the welder concerned. The cable jacket shall be natural rubber, or styrene butadiene rubber (SBR) or polyethylene black material.

3.8.3 High frequency and gas control. High frequency and gas controls shall be built-in or an attachment as specified (see 6.2.1). The built-in high frequency unit shall have sufficient capacity to provide high frequency for welding up to a distance of 40 feet from the power source, and the gas control unit shall have automatic gas control and adjustable post-flow timer. The high frequency attachment shall be compatible to the power source so that the attachment can operate at a distance up to 40 feet from the power source and provide high frequency for welding up to a distance of 20 feet from the attachment.

3.9 Nameplate. A nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the welding machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Engine make, model, HP
AC and DC amp rating at rated volts; duty cycle,
  AC and DC amp range
Auxiliary electric power KW, voltage
Date of manufacture
Contract or order number
National Stock Number or Plant Equipment Code

3.10 Hour meter. Unless otherwise specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the engine. The meter shall be of the nonresetting type and shall have a range of 0 to 9999.9 hours. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

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3.11 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.12 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the item to pass the first article tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of at least the examination in 4.5 and the tests in 4.6.1. Failure of the item to pass the examination or tests shall be cause for rejection.

4.5 Examination. The machine and equipment shall be examined to determine compliance with the requirements specified herein.

4.6 Tests.

4.6.1 Operational test. The machine shall be operated at no load until the engine temperature stabilizes at operating temperature. Using a resistive load, the output of the welding machine shall be checked through its operating range. All controls and instruments shall be checked. The machine shall comply with all requirements specified herein.

4.6.2 High potential test. Each welding machine shall be subjected to the high potential test described in "Testing Standards" of EW-1. When first article inspection is required, this test is to be performed after test 4.6.3.

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4.6.3 Rough handling test. A chain or cable shall be attached to the welding machine lifting eyes. The chain or cable shall be adjusted in length to permit a "free fall" of at least six inches and not more than ten inches before the machine is suspended thereby. The machine shall survive three such falls without damage to the effect that will impair the performance of the machine. Additionally, the machine shall be dropped a distance of six inches to a rigid, hard surface. Dropping shall be arranged so that the initial impact is successively made on each of three of the lower edges. No measureable damage, other than minor denting shall be encountered. This test shall be performed prior to all other tests when first article inspection is required.

4.6.4 Performance test. To demonstrate the capability of the machine to weld within its specified range, a skilled operator shall deposit six-inch long weld beads using AC and DC with both polarities. Welding shall be done with 1/8 and 1/4-inch diameter electrodes applicable for the processes. There shall be no evidence of difficulty in striking or maintaining the arc.

4.6.5 Emission test. When specified (see 6.2.1), the engine shall be run at idling speed and no load and the exhaust tested to determine compliance with emission requirements. The test shall then be repeated with the engine at full load and operating speed. Exhaust emissions shall be within limits allowed by Government regulations.

4.6.6 Parking brake test. The fully loaded trailer shall be placed on a 15 degree incline and the parking brake applied. The engine of the welding machine shall be started and run at normal operating speed. There shall be no movement of the trailer down the incline under the above conditions.

4.6.7 Trailer road test. When an over the road type trailer is required it shall be tested for roadability. The fully loaded trailer shall be towed for 10 miles at approximately 50 mph on improved highways and 10 miles at approximately 10 mph on rough terrain. During this test, the trailer shall track steady, with minimum side sway, and cause no difficulty in handling the towing vehicle. At the end of this test, the welding machine shall be examined for damage due to road shock. Failure to track steady, difficulty in handling the towing vehicle, or damage due to road shock shall be cause for rejection.

4.7 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging, packing and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM-D-3951. When specified, preservation, packaging, packing and marking shall be in accordance with the applicable requirements of MIL-W-45562. Level of preservation, packaging and packing shall be as specified (see 6.2.1).

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6. NOTES

6.1 Intended use. The machines covered by this specification are to provide an AC/DC power source where other electric power is not available; where portability is desired in shops, field maintenance and construction; and where auxiliary electric power is desired in connection with the welding operation.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required, (see 1.2).

c. When first article is required for inspection and approval (see 3.1).

d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. Dual calibrations (US and SI), if required (see 3.2.1.1).

f. Additional safety and health requirements, if required (see 3.2.4).

g. Painting, if different (see 3.3.5).

h. Batteries, if required (see 3.4.1.2).

i. Electrical starting system (type I machine), if required (see 3.4.1.2).

j. Low oil pressure shutdown, if required (see 3.4.1.3).

k. Excessive temperature shutdown device, if required (see 3.4.1.10).

l. Automatic idling device, if required (see 3.4.1.11).

m. Fuel gage, if required (see 3.4.2.2).

n. Two hundred thirty-volt receptacle and KW output, if required (see 6 3.4.1.11).

o. Machine mounting, if different (see 3.7).

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p. Over the road trailer, if required (see 3.7.1).

q. Spare tire for trailer, if required (see 3.7.1).

r. Welding cable length, specify (see 3.8.2).

s. High frequency gas control, (built-in or attached) specify (see 3.8.3).

t. Quality conformance inspection, if different (see 4.4).

w. Engine emission test, if required (see 4.6.5).

x. Level of packaging, packing and marking, if different (see 5.1).

6.2.2 Contract data requirements. Required technical data, such as, operator's manuals, parts lists, wiring diagrams, examination, alignment and tolerance test reports and other instructions for operation and maintenance (as identified on a DD Form 1664, Data Item Description), should be specified on a DD Form 1423, Contract Data Requirements List incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine it its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total hazard level of individual machines in the environment. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1(f).

Custodians:
Army - ME
Navy - YD
Air Force - 99

Review Activities:
Army - ME, AL
Navy - SH, YD
Air Force - 84
DLA - GS

Using Activities:
Navy - YD, MC, OS
Army - CE

Preparing Activity:
DLA-IP

Project Number:
3431-0156

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-80080C	Welding Machines, Arc, AC/DC, Generator, Constant Current, Internal Combustion Engine Driven

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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